

4E1305

Roll No. _____

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B.Tech. IV Sem. (Main) Examination, - July 2023
Computer Science and Engineering (AI)
4CAI4-05 Database Management System
CS, IT, AID, CAI

Time : 3 Hours**Maximum Marks : 70****Instructions to Candidates:**

Attempt all ten questions from Part A, Attempt any five questions out of Seven questions from Part B and three questions out of Five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (As Mentioned in form No. 205).

PART - A**(Word limit 25)****All questions are compulsory****(10×2=20)**

1. Write any four differences between file system and DBMS.
2. Draw state diagram of transaction.
3. What is ODBC? How does it differ from JDBC?
4. Consider a relation R(A,B,C,D,E) with A,B,C,D,E as attributes and functional dependency set $F = \{AB \rightarrow C, C \rightarrow D, B \rightarrow E\}$. Find out the candidate key.
5. How does weak entity differ from strong entity in ER model?
6. Explain various levels of data abstractions in DBMS.
7. How does correlated nested queries differ from simple nested queries. Explain with example.
8. Explain Aggregation with example.
9. Differentiate Generalization and Specialization.
10. In which applications, Embedded and Dynamic SQL are required?

PART - B
(Word limit 100)

Attempt any Five questions.

(5×4=20)

1. In a class of 60 students, 25 study Hindi, 26 study English and 26 study Sanskrit. Also 9 study both Hindi and Sanskrit, 11 study hindi and english and 8 study Englishh and Sanskrit. If 8 study none of the three subjects, Find number of students who study exactly one subject. (4)
2. Using principle of mathematical induction, prove that $1 + 2 + 2^2 + \dots + 2^{n-1} = 2^n - 1$. (4)
3. Draw the transition diagram of finite state machine represented by following state table. Also find output word corresponding to input word $w = 11011011$. (4)

State	Transition	function	Output	function
	0	1	0	1
s_0	s_3	s_1	0	1
s_1	s_0	s_1	0	1
s_2	s_3	s_1	0	1
s_3	s_1	s_3	0	0

4. Solve the recurrence relation

$$a_n - 4a_{n-1} + 4a_{n-2} = 0 ; a_0 = 1, a_1 = 3. \quad (4)$$

5. Define isomorphism of groups. Prove that every subgroup of an abelian group is normal. (4)
6. Prove that the sum of degrees of all the vertices in a graph is equal to twice the number of edges in the graph. (4)
7. Prove that set of real numbers of the form $a + b\sqrt{2}$ (where a and b are integers) with ordinary addition and multiplication forms a ring. (4)

PART - C

Attempt any Three questions.

(3×10=30)

1. In the set Z of integers, a relation R is defined by $aRb \Leftrightarrow a \equiv b \pmod{4}$. Show that R is an equivalence relation. (10)
2. Define universal and existential quantifiers. prove that $p \rightarrow (q \wedge r) = (p \rightarrow q) \wedge (p \rightarrow r)$. (10)
3. Prove that the dual of a lattice is also a lattice. (10)

4. Prove that the necessary and sufficient conditions for a non - void subset H of a group G to be a subgroup is that $a \in H, b \in H \Rightarrow ab^{-1} \in H$. (10)
5. Find the shortest path between the vertices v_0 and v_1 in the following weighed graph. (10)

